

Patrick Woodburn

pat@patrickwoodburn.com | (267) - 231 - 2069 | Philadelphia, US

Summary

A jack of all trades engineer with more than a decade of experience delivering software at scale. I have a passion for learning and applying new technologies to solve problems. I have a strong background in software engineering, but I also have experience in electrical design and prototyping. I am always looking for new challenges and opportunities to grow as an engineer.

Skills

- **Back End Services:** Java, spring boot, REST, Protobuf, gRPC, Redis, DynamoDB, PostgreSQL
- **Embedded Linux:** C/C++, CMake, yacto, syscalls, systemd
- **DevOps:** git, github, gitlab, CI/CD, ci pipelines, Jenkins, Docker, ansible, terraform, aws
- **Microcontrollers:** Arduino, PIC, AVR, Arm, BLE, C/C++
- **Electrical Design:** Circuit Design, Prototyping, soldering, CAD, PCB design, 3D printing

Work Experience

Syzygy Integration

Senior Software Engineer

Jun 2026 — Present

- Create and maintain embedded services for communications between edge devices and embedded sensors
- Handle integration of new sensors and devices into the Syzygy Integration platform
- Push towards modernizing development practices and tools to increase developer productivity and reduce time to market for new features

Samsung SmartThings

Senior Software Engineer

Aug 2022 — Mar 2026

- Created and maintained microservices that run on AWS to support the SmartThings platform.
- Used Terraform to manage infrastructure as code for the SmartThings platform.
- Optimized performance of jvm and ec2 instances to reduce costs and increase reliability of the SmartThings platform.
- Leveraged agentic workflows to accelerate api migration to newer versions of Java and more modern frameworks.
- Maintained api documentation for services to ensure that they are up to date and accurate for both internal and external use.
- Design Dynamo Tables for optimal performance and cost efficiency as well as ease of use for developers.
- Enhanced monitoring and alerting systems to ensure that issues with the SmartThings platform are quickly identified and resolved.
- Handled on call responsibilities to maintain the reliability and uptime of the SmartThings platform.

Fort Robotics

Senior Software Engineer

Jun 2021 — Jul 2022

- Worked with yocto to create embedded linux distributions for custom hardware.
- Created standard build environments to enable developers to build consistent software.
- Increased performance of CMake build tools.
- Created chip flashing tools for low level radio interfaces.
- Worked on embedded communications system using zmq and coap.
- Set up automatic build systems for multiple code repositories in multiple programming languages.
- Improved code stability through the implementation of static code analysis.
- Transitioned dependency management to more manageable system allowing for faster more reliable software builds.
- Worked on embedded hardware to increase safety in industrial environments.

Thomas Jefferson University - Dice Group

Software Engineer

Dec 2017 — Jun 2021

- Worked on software that drove the first ever HIPPA compliant smart speaker.
- Built APIs that allowed for internet connected devices to communicate between each other for the purpose of controlling equipment in a hospital setting.
- Wrote Linux scripts used to manage a suite of software focused on the embedded platform.
- Built a system for intercommunication between APIs and firewall protected embedded systems.
- Designed a system for delivering software updates leveraging MQTT events.
- Used Google Cloud Platform to control and maintain numerous Jefferson IoT projects.
- Improved data flow between existing APIs to ensure negligible delay between interconnected systems.
- Lead conversations on uses, pros and cons, and implementation of IoT networks including LoRaWAN, Bluetooth, WiFi, and cellular within the engineering team.
- Designed and created prototypes of simple electrical circuits for small batch products on experimental IoT and embedded system projects.
- Designed 3D printed short run rapid prototype housings for early phase projects during the proof of concept stage.
- Created an interface for controlling Docker over the MQTT network.
- Maintained Docker images for testing embedded software in order to eliminate the need for physical hardware.
- Created workflows for using Docker on embedded systems.
- Created systems to automatically and dynamically deploy software to embedded hardware.
- Implement Redis as a low lag reliable cross application key store and communication bus solution.

Comcast

Software Engineer

Mar 2015 — May 2017

- Operated within a team of developers to develop and test a new home automation system.
- Worked with internet connected devices to automate and secure user's homes.
- Developed android apps that could run against a users unique home automation system.
- Used testing tools such as junit and espresso to create test code to run against android apps for continuous integration.
- Built test kits to simulate devices in a typical user's home.
- Created technical documentation for both installation technicians and tech support to use when assisting users.
- Set up continuous integration environment using Jenkins.
- Scripted tools to deploy to and debug on custom android firmware.
- Developed software to communicate over wireless interfaces such as bluetooth, WIFI, and Zigbee.

Franklin Institute

Science Educator

Apr 2006 — Jun 2019

- Planned lessons on the topics of electrical and software development.

- Taught students how to program and build circuitry required for multiple applications of robotics.
- Worked with students to build robots for various competition and museum displays.
- Documented use and maintenance of students' projects to be handed off to groups tasked with operation through museum.
- Built software tools that the students can use to better debug their projects.
- Presented and demonstrated robotics projects to large groups of people.
- Operated robots in and around the museum.

Independence Blue Cross

Software Engineer

Aug 2014 — Feb 2015

- Implemented given designs so that users may have the most comfortable experience possible when accessing their insurance information.
- Created cross-platform mobile apps that could be rebranded for multiple insurance companies.
- Ported functionality of apps between incompatible code repositories.

BioHitech America

Intern

Jan 2014 — Jul 2014

- Developed software to run on headless industrial machinery and perform analytics regarding their use.
- Set up and maintained Linux devices to be used in embedded industrial applications.
- Created static environment configuration for linux machinery and virtual machines.
- Developed Kiosk software to be able to track metrics from items scanned with nfc chips.
- Developed webpages to pull from various RESTful APIs.
- Created firmware to drive touchscreen pannels.
- Researched and retrofitted new hardware for industrial-grade machinery.

Volunteering

Code for Philly

May 2011 — Jan 2016

Developed Open source software for the betterment of Philadelphia.

- Acted as a subject matter expert on developing API's and integrating them into web and mobile apps.
- Created websites for community service projects.
- Built apps to help transform improve communities within the city.
- Developed APIs to open up public data from the city.
- Helped others to bring up and configure their server solutions on a project to project basis.
- Built data scrapers to bring data into databases from other sources.

Education

Harrisburg University Of Science And Technology

Bachelor of Science in Computer Information Sciences

Sep 2010 — May 2014